

UC Berkeley
Teaching Professor
Dan Garcia

CS61C

Great Ideas in Computer Architecture

(a.k.a. Machine Structures)



UC Berkeley
Teaching Professor
Lisa Yan

Systems Lunch on the Glade!
Friday, October 7, 12pm
Dan, Lisa, and
Prof. Natacha Crooks (CS162)
Bring your own lunch
and come join us!



RISC V Datapath I

Berkeley
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cs61c.org

Garcia, Yan

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Building a RISC-V Processor

- Building a RISC-V Processor
- CPU Elements and Stages
- R-Type: `add` datapath
- R-Type: `sub` datapath
- Datapath with immediates: `addi`

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Great Idea #1: Abstraction (Levels of Representation/Interpretation)

High Level Language Program (e.g., C)

Compiler

Assembly Language Program (e.g., RISC-V)

Assembler

Machine Language Program (RISC-V)

```
temp = v[k];
v[k] = v[k+1];
v[k+1] = temp;
```

```
lw    x3, 0(x10)
lw    x4, 4(x10)
sw    x4, 0(x10)
sw    x3, 4(x10)
```

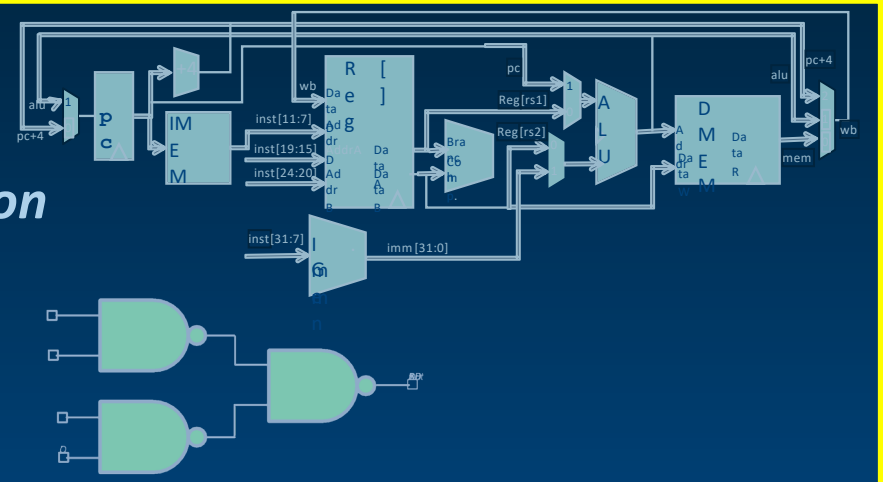
```
1000 1101 1110 0010 0000 0000 0000 0000
1000 1110 0001 0000 0000 0000 0000 0100
1010 1110 0001 0010 0000 0000 0000 0000
1010 1101 1110 0010 0000 0000 0000 0100
```



Hardware Architecture Description (e.g., block diagrams)

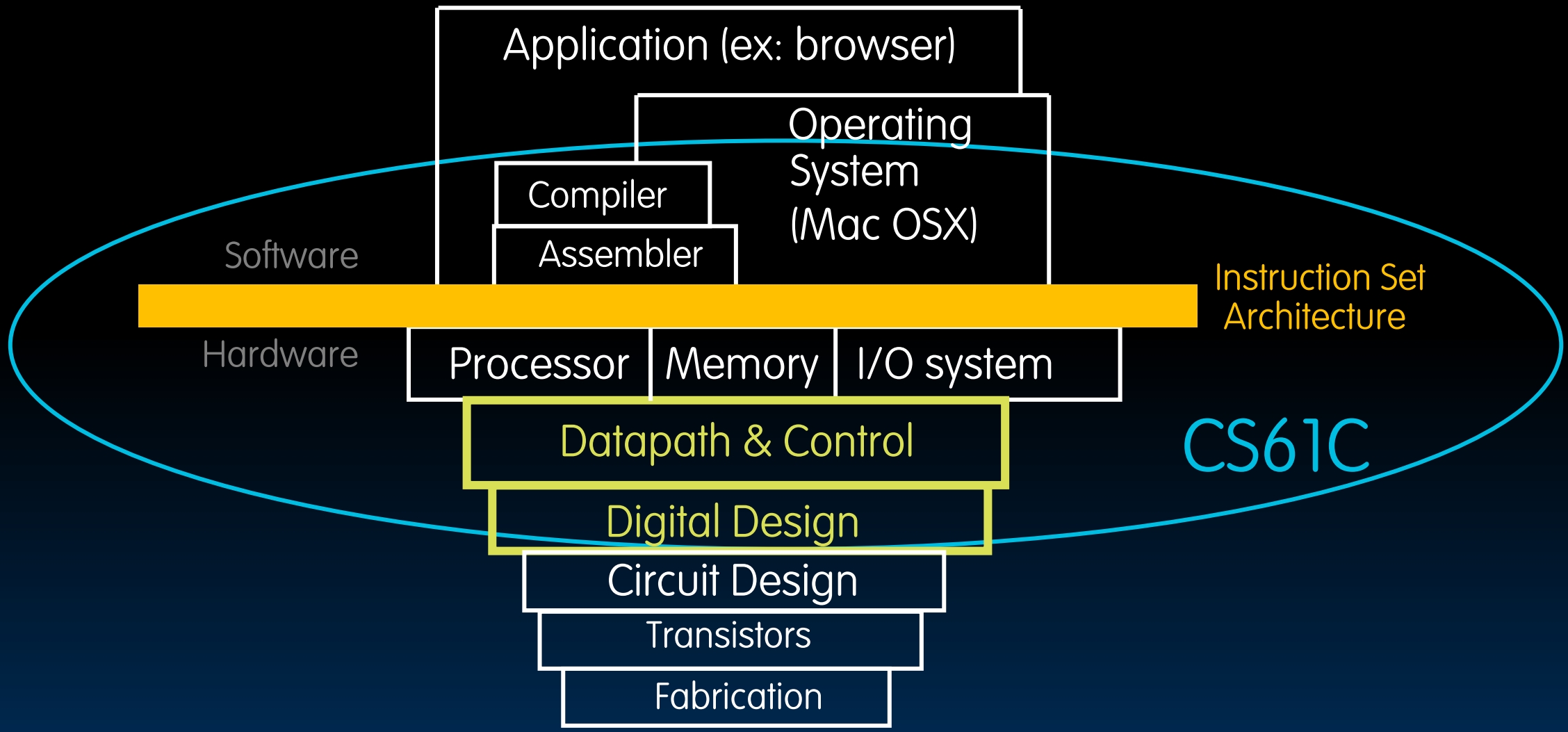
Architecture Implementation

Logic Circuit Description (Circuit Schematic Diagrams)



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Old-school Machine Structures



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New-school Machine Structures

Software

Harness Parallelism & Achieve High Performance

Parallel Requests
Assigned to computer
e.g., Search "Cats"

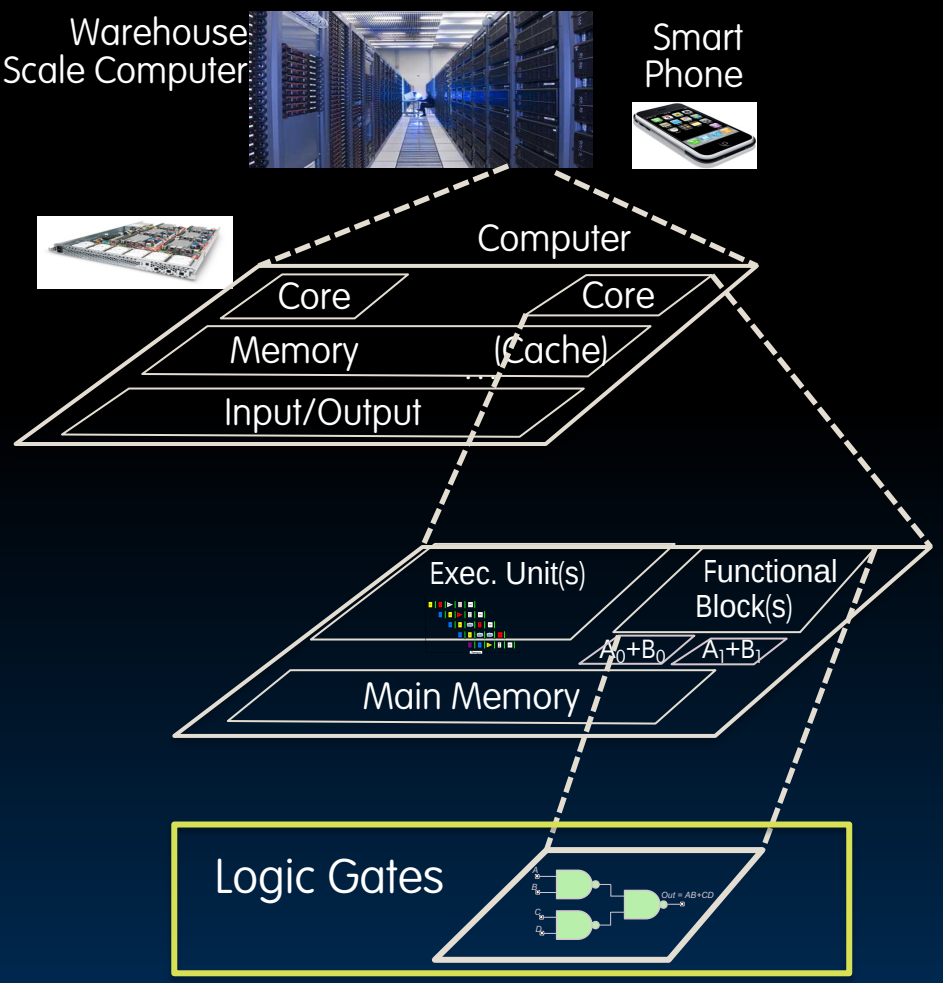
Parallel Threads
Assigned to core e.g., Lookup, Ads

Parallel Instructions
>1 instruction @ one time
e.g., 5 pipelined instructions

Parallel Data
>1 data item @ one time
e.g., Add of 4 pairs of words

Hardware descriptions
All gates work in parallel at same time

Hardware

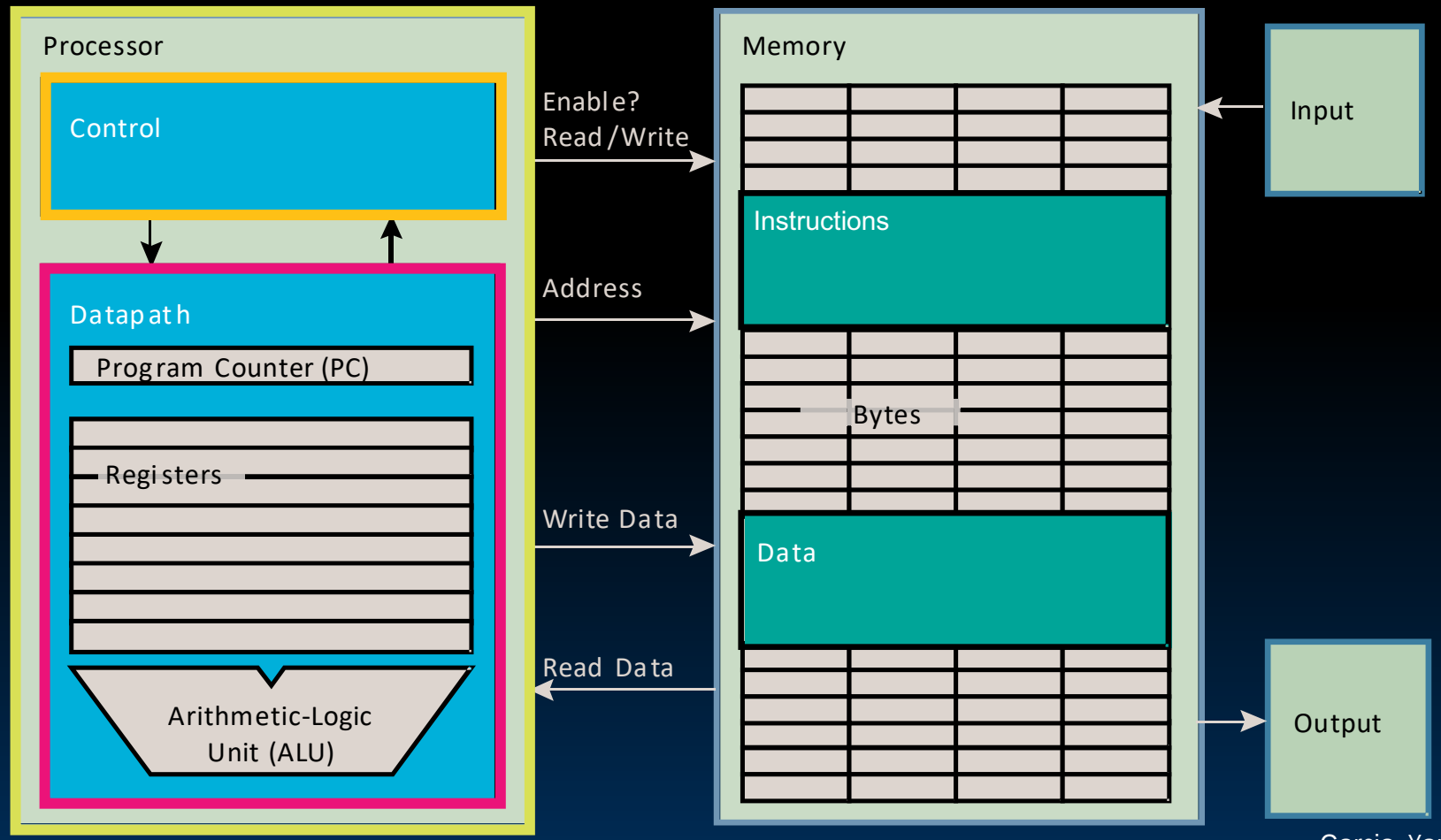


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How do we build a Single-Core Processor?

Processor (CPU): the active part of the computer that does all the work (data manipulation and decision-making).

- *Datapath* ("the brawn"): portion of the processor that contains hardware necessary to perform operations required by the processor.
- *Control* ("the brain"): portion of the processor (also in hardware) that tells the datapath what needs to be done.



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Goal: Design HW to Execute All RV32I Instructions

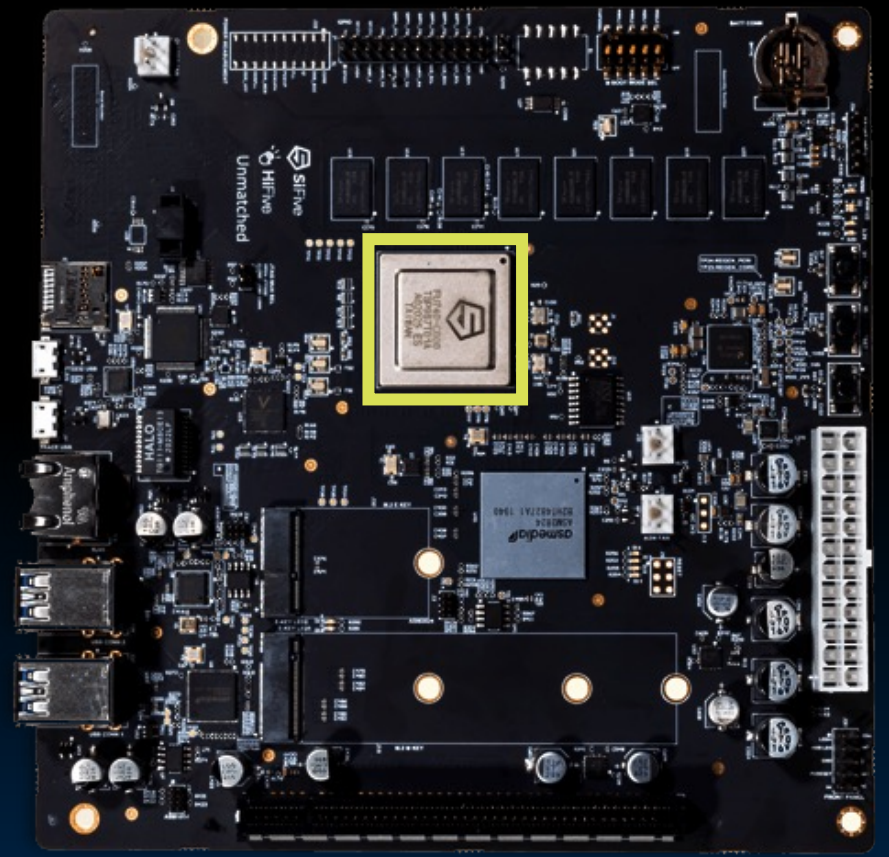
Open RISC-V Reference Card

Base Integer Instructions: RV32I

Category	Name	Fmt	RV32I Base
Shifts	Shift Left Logical	R	SLL rd,rs1,rs2
	Shift Left Log. Imm.	I	SLLI rd,rs1,shamt
	Shift Right Logical	R	SRL rd,rs1,rs2
	Shift Right Log. Imm.	I	SRLI rd,rs1,shamt
	Shift Right Arithmetic	R	SRA rd,rs1,rs2
	Shift Right Arith. Imm.	I	SRAI rd,rs1,shamt
Arithmetic	ADD	R	ADD rd,rs1,rs2
	ADD Immediate	I	ADDI rd,rs1,imm
	SUBtract	R	SUB rd,rs1,rs2
	Load Upper Imm	U	LUI rd,imm
	Add Upper Imm to PC	U	AUIPC rd,imm
Logical	XOR	R	XOR rd,rs1,rs2
	XOR Immediate	I	XORI rd,rs1,imm
	OR	R	OR rd,rs1,rs2
	OR Immediate	I	ORI rd,rs1,imm
	AND	R	AND rd,rs1,rs2
	AND Immediate	I	ANDI rd,rs1,imm
Compare	Set <	R	SLT rd,rs1,rs2
	Set < Immediate	I	SLTI rd,rs1,imm
	Set < Unsigned	R	SLTU rd,rs1,rs2
	Set < Imm Unsigned	I	SLTIU rd,rs1,imm

Category	Name	Fmt	RV32I Base
Loads	Load Byte	I	LB rd,rs1,imm
	Load Halfword	I	LH rd,rs1,imm
	Load Byte Unsigned	I	LBU rd,rs1,imm
	Load Half Unsigned	I	LHU rd,rs1,imm
	Load Word	I	LW rd,rs1,imm
Stores	Store Byte	S	SB rs1,rs2,imm
	Store Halfword	S	SH rs1,rs2,imm
	Store Word	S	SW rs1,rs2,imm
Branches	Branch =	B	BEQ rs1,rs2,imm
	Branch ≠	B	BNE rs1,rs2,imm
	Branch <	B	BLT rs1,rs2,imm
	Branch ≥	B	BGE rs1,rs2,imm
	Branch < Unsigned	B	BLTU rs1,rs2,imm
	Branch ≥ Unsigned	B	BGEU rs1,rs2,imm
Jump & Link	J&L	J	JAL rd,imm
	Jump & Link Register	I	JALR rd,rs1,imm
Synch	Synch thread	I	FENCE
Environment	CALL	I	ECALL
	BREAK	I	EBREAK

Not in 61C



HiFive Unmatched (RISC-V Linux)

https://www.sifive.com/boards/hifive-unmatched

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CPU Elements and Stages

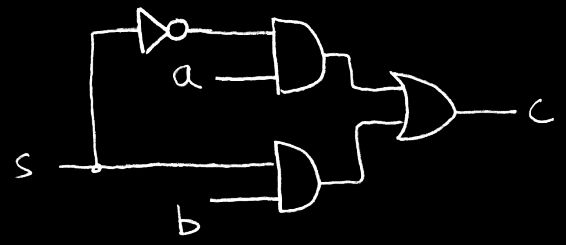
- Building a RISC-V Processor
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- R-Type: `add` datapath
- R-Type: `sub` datapath
- Datapath with immediates: `addi`

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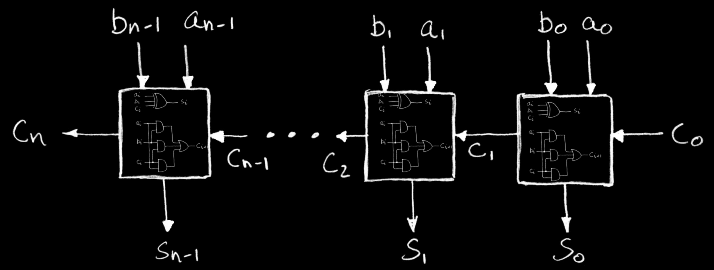
Combinational Logic Blocks

- **Today, we will think of all combinational logic subcircuits as block diagrams:**

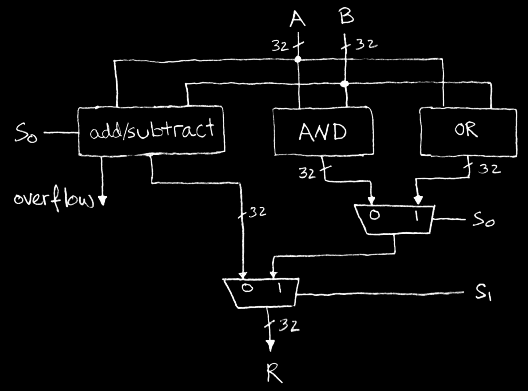
Last time



1-bit-wide 2-to-1MUX

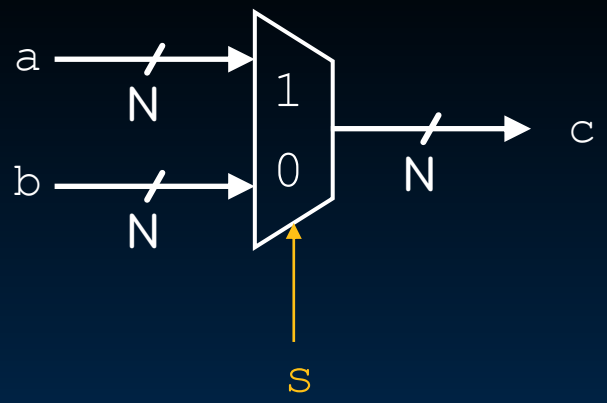


N-Bit Adder

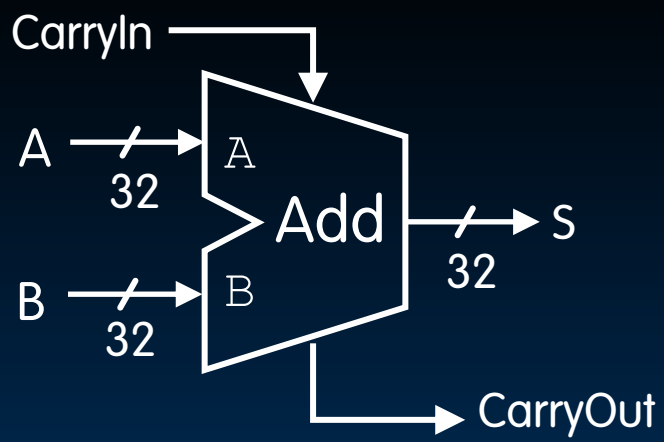


Simple ALU (add,sub,and,or)

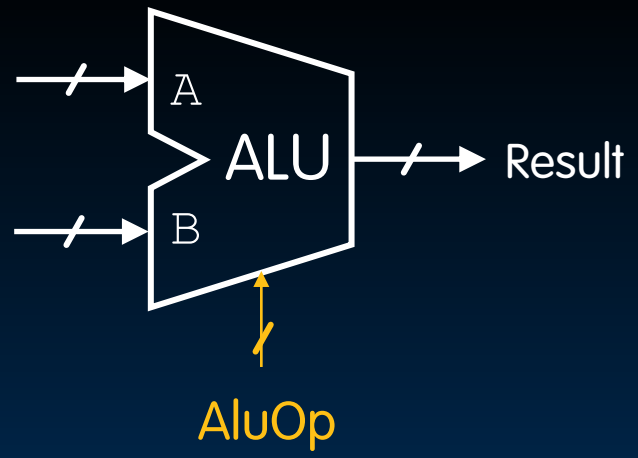
This time



32-bit-wide 2-to-1 MUX



32-Bit Adder



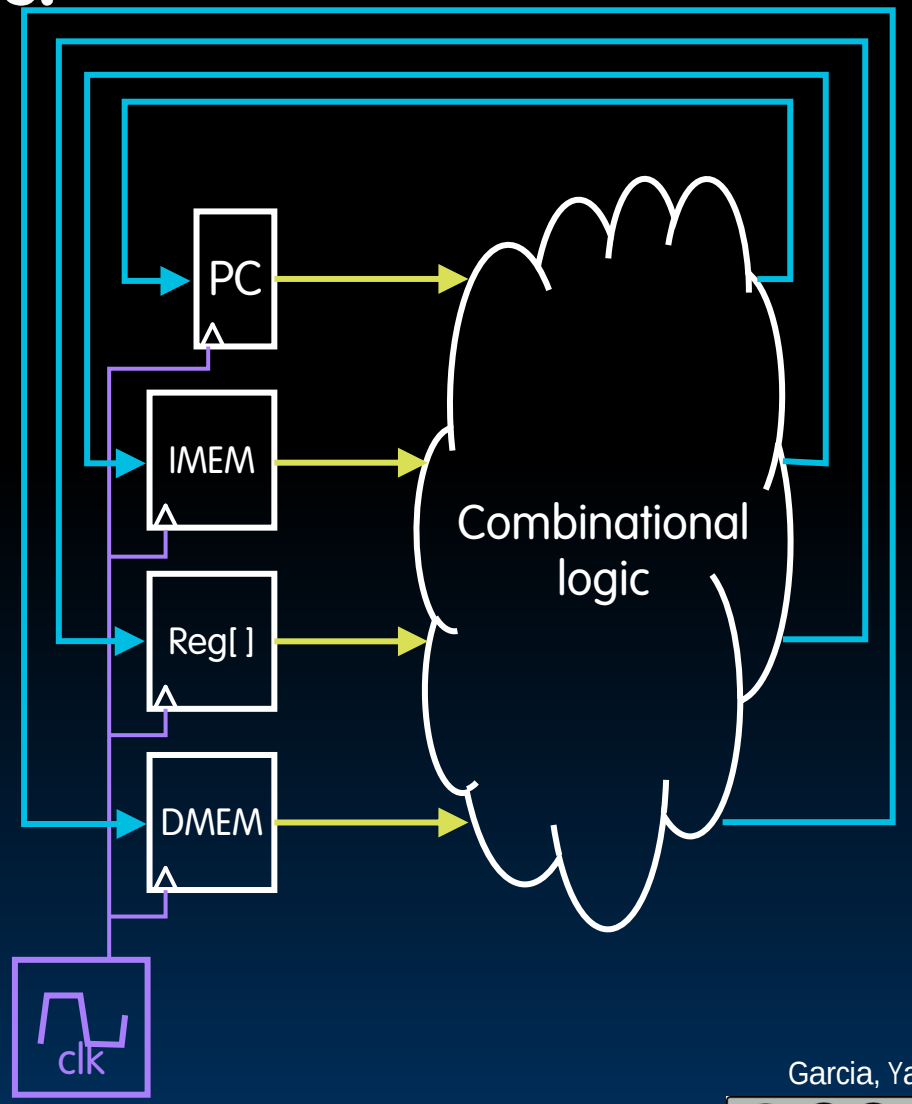
ALU (ALUOp selects from multiple operations)

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One-Instruction-Per-Cycle RISC V Machine

The CPU is composed of two types of subcircuits: *combinational logic blocks* **and** *state elements*.

- **On every tick of the clock, the computer executes one instruction:**
  - Current outputs of the *state elements* drive the inputs to combinational logic…
  - …whose outputs settle at the inputs to the state elements before the next rising clock edge.
- **At the rising clock edge:**
  - All the *state elements* are updated with the combinational logic outputs…
  - and execution moves to the next clock cycle.



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State Elements Required by RV32I ISA

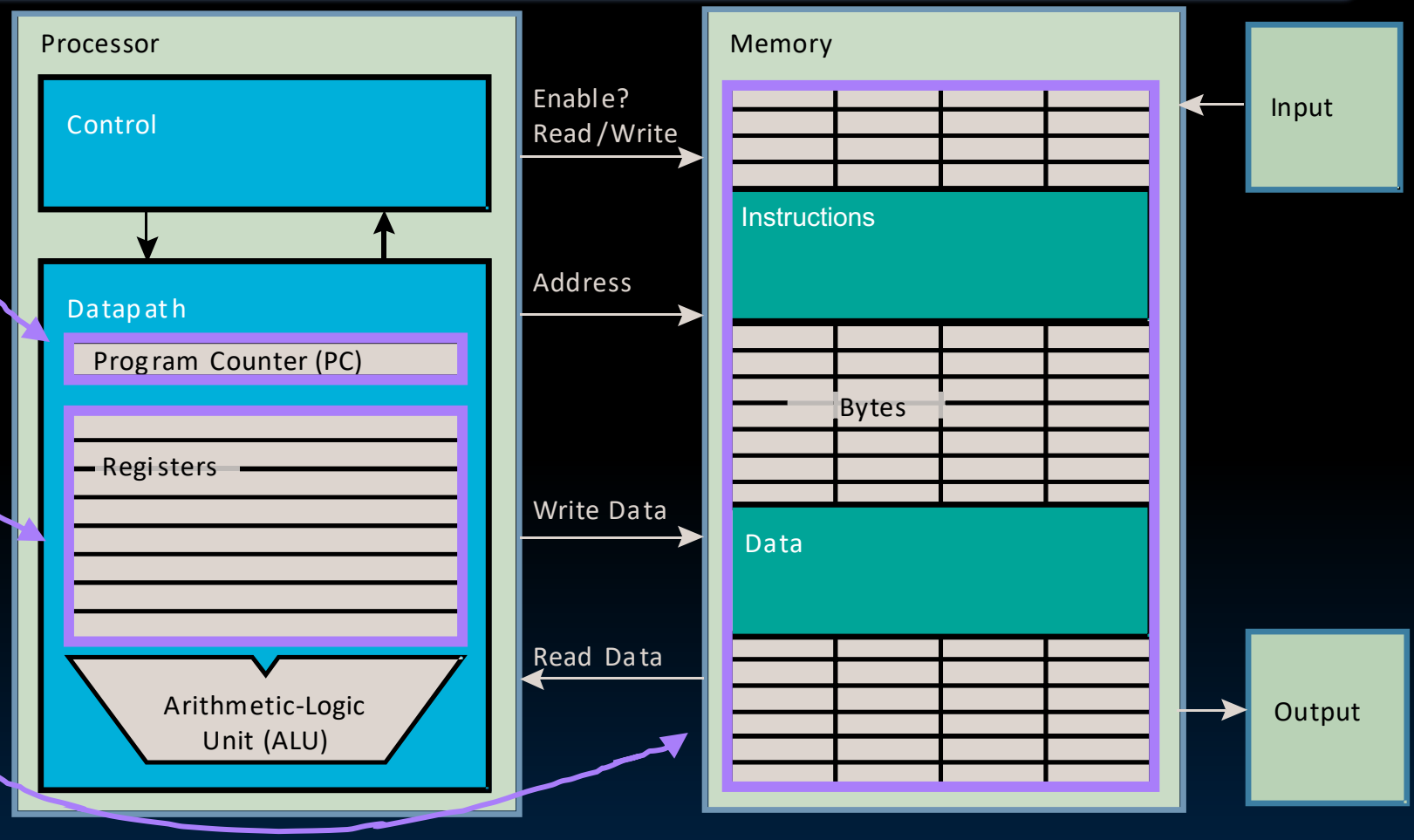
Program Counter PC

Register File Reg[ ]

Memory MEM

IMEM

DMEM

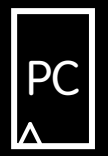


During CPU execution, each RV32I instruction reads and/or updates these state elements.

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State Elements (1/3): Program Counter

Program Counter



Register File Reg

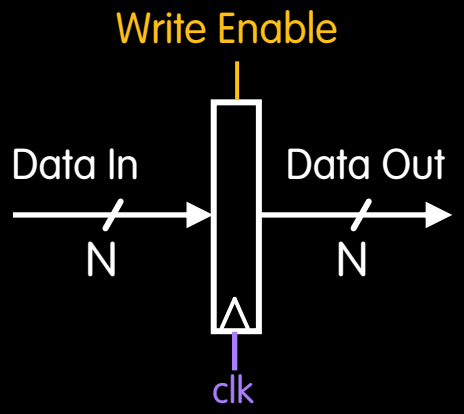


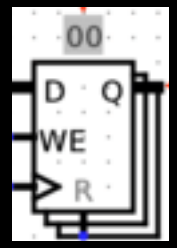
Memory MEM











A register in Logisim

The Program Counter is a 32-bit Register.

- Input:
  - N-bit data input bus
  - Write Enable "Control" bit (1: asserted/high, 0: deasserted/0)
- Output:
  - N-bit data output bus
- Behavior:
  - If Write Enable is 1 on *rising clock edge*, set Data Out=Data In.
  - *At all other times*, Data Out will not change; it will output its current value.

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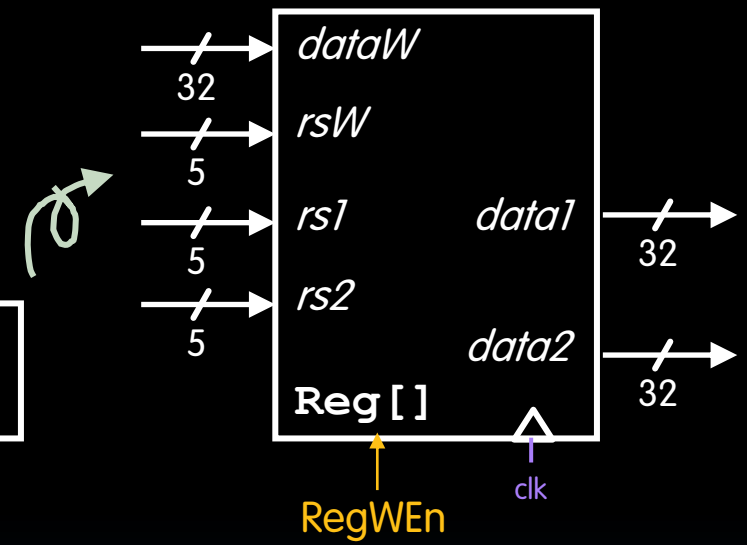
State Elements (2/3): Register File

Program Counter

PC

Register File Reg

Reg[ ]



Register File (RegFile) has 32 registers.

- Input:
  - One 32-bit input data bus, *dataW*.
  - Three 5-bit select busses, *rs1*, *rs2*, and *rsW*.
  - RegWEn control bit.
- Output:
  - Two 32-bit output data busses, *data1* and *data2*.

Memory MEM

IMEM

DMEM

- Registers are accessed via their 5-bit register numbers:
  - `R[rs1]`: *rs1* selects register to put on *data1* bus out.
  - `R[rs2]`: *rs2* selects register to put on *data2* bus out.
  - `R[rd]`: *rsW* selects register to be written via *dataW* when RegWEn=1.
- Clock behavior: Write operation occurs on *rising clock edge*.
  - Clock input only a factor on write!
  - All read operations behave like a combinational block:
    - If *rs1*, *rs2* valid, then *data1*, *data2* valid after *access time*.

RegFile behaves like a combinational block for read operations!

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State Elements (3/3): Memory

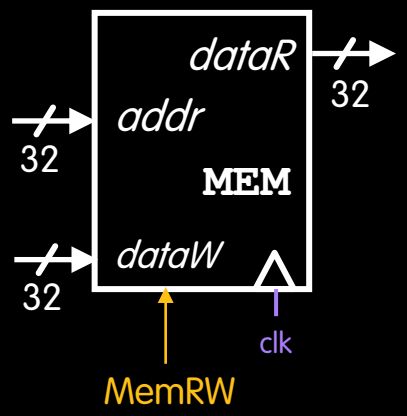
Program Counter PC

Register File Reg[ ]

Memory MEM

IMEM

DMEM



Memory is "magic." For this class:

- 32-bit byte-addressed memory space; and
- Memory access with 32-bit words.

- Memory words are accessed as follows:
  - Read: Address *addr* selects word to put on *dataR* bus.
  - Write: Set MemRW=1.
    Address *addr* selects word to be written with *dataW* bus.
- Like RegFile, clock input is only a factor on write.
  - If MemRW=1, write occurs on rising clock edge.
  - If MemRW=0 and *addr* valid, then *dataR* valid after access time.

If MemRW=0, **MEM** behaves like a combinational block.

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"Two" Memories: IMEM, DMEM

Program Counter



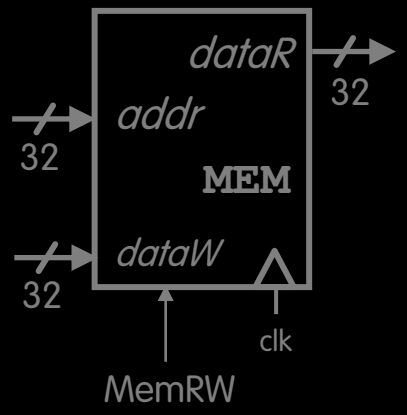
Register File Reg



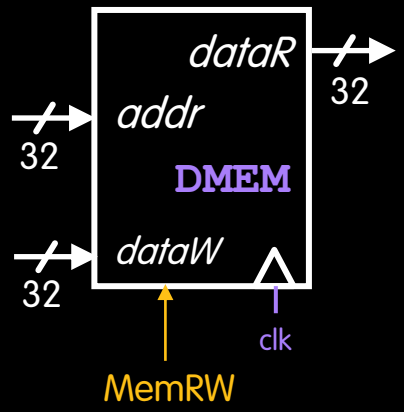
Memory MEM









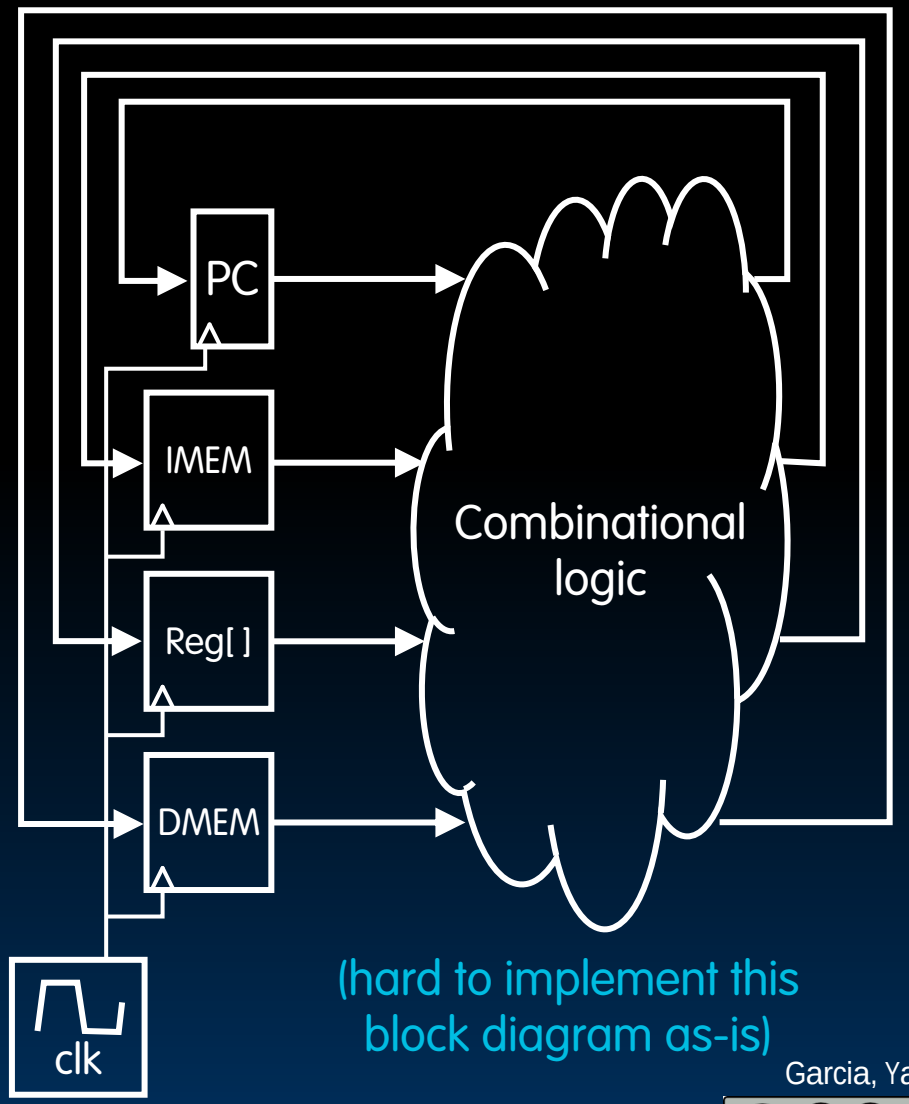


- Current abstraction: Memory holds both instructions and data in one contiguous 32-bit memory space.
- In our processor, we'll use two "separate" memories:
  - **IMEM**: A *read-only* memory for fetching instructions.
  - **DMEM**: A memory for loading (read) and storing (write) data words.
  - Under the hood, these are placeholders for caches. (more later)
- Because IMEM is read-only, it always behaves like a combinational block:
  - If *addr* valid, then *instr* valid after access time.

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Design the Datapath in Phases

- **Task: "Execute an instruction."**
  - All necessary operations, starting with fetching the instruction.
- Problem: **A single "monolithic" block would be bulky and inefficient.**
- Solution: **Break up the process into stages, then connect the stages to create the whole datapath.**
  - Smaller stages are easier to design!
  - *Modularity*: Easy to optimize one stage without touching the others.

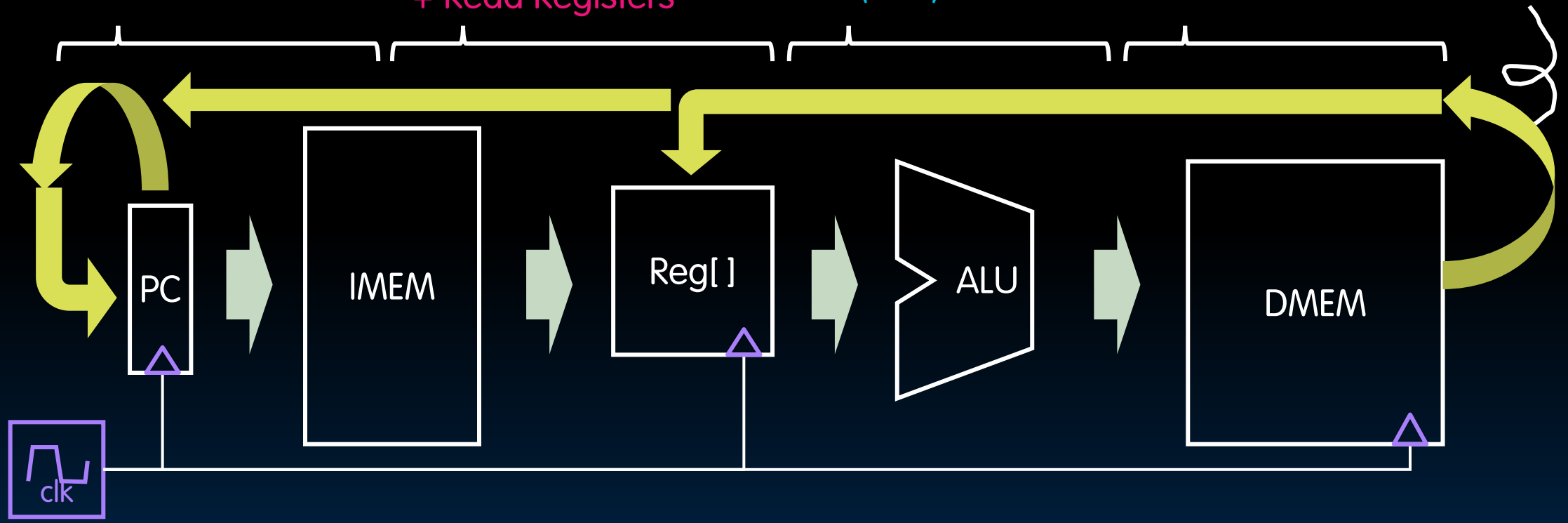


(hard to implement this block diagram as-is)

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5 Basic Stages (Phases) of Instruction Execution

1. Instruction Fetch (`IF`)
2. Instruction Decode (`ID`) + Read Registers
3. Execute (`EX`) Arithmetic Logic Unit (ALU)
4. Memory Access (`MEM`)
5. Write back to Register (`WB`)

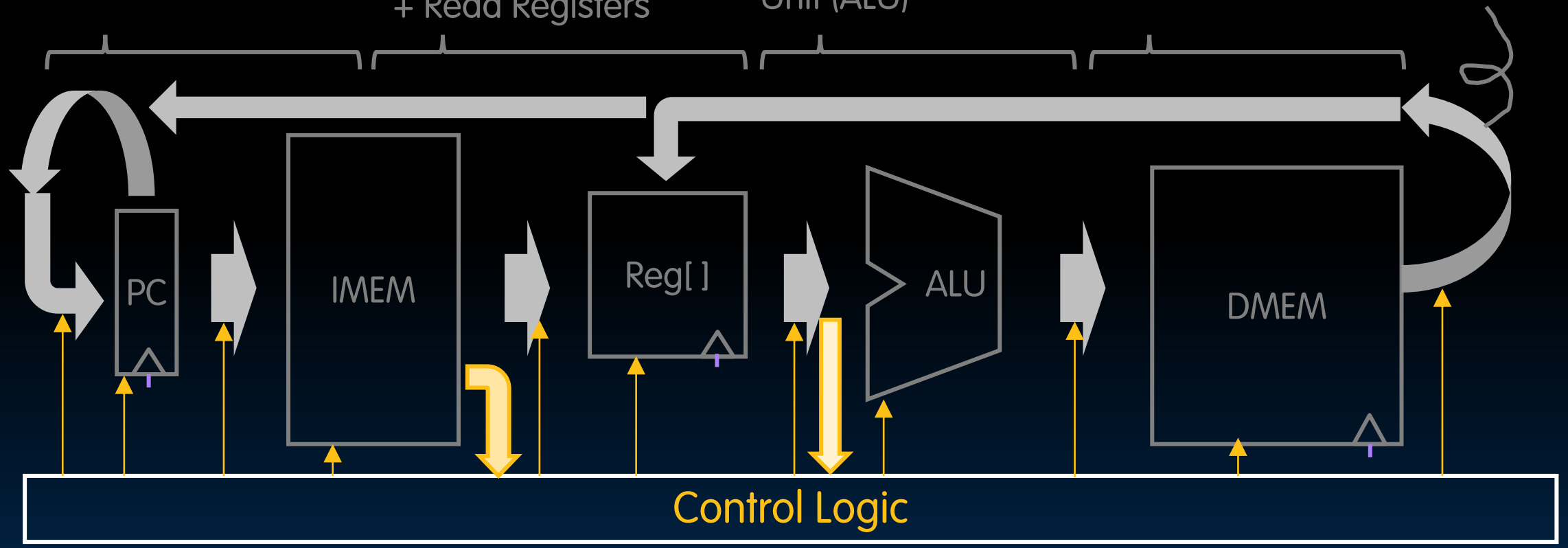


- **We will implement a single-cycle processor:**
  - All stages of one RV32I instruction execute within the same clock cycle.

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Not All Instructions Need All 5 Stages!

1. Instruction Fetch (**IF**)
2. Instruction Decode (**ID**) + Read Registers
3. Execute (**EX**) Arithmetic Logic Unit (ALU)
4. Memory Access (**MEM**)
5. Write back to Register (**WB**)



The control logic selects "needed" datapath lines based on the instruction.

- MUX selector, ALU op selector, write enable, etc.

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R-Type: `add` Datapath

- Building a RISC-V Processor
- CPU Elements and Stages
- R-Type: **`add`** datapath
- R-Type: **`sub`** datapath
- Datapath with immediates: **`addi`**

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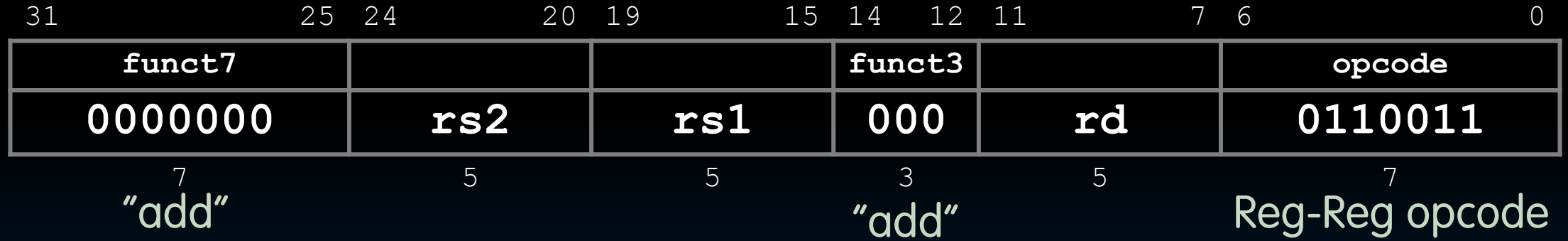
Implementing the add Instruction

- **Suppose we had a single instruction in our RISC-V ISA: add.**

Example add-only program

0x100	add x18,x18,x10
0x108	add x18,x18,x18
…	add …

add rd, rs1, rs2



- **The add instruction makes two changes to processor state:**
  - RegFile
  - PC

```
Reg[rd] = Reg[rs1] + Reg[rs2]
PC = PC + 4
```


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Datapath for add

31 25	24 20	19 15	14 12	11 7	6 0
funct7	rs2	rs1	funct3	rd	0110011

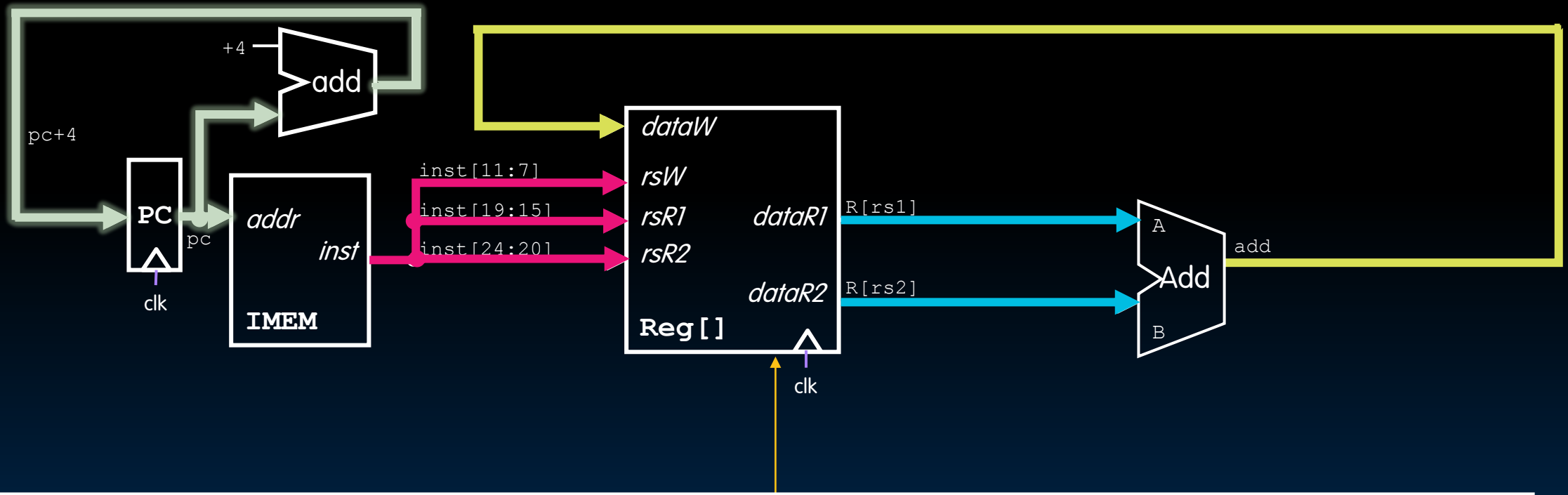
```
PC = PC + 4          R[rd] = R[rs1] + R[rs2]
```

Increment PC to next instruction.

Split instruction to index into RegFile.

Feed read register values into Add.

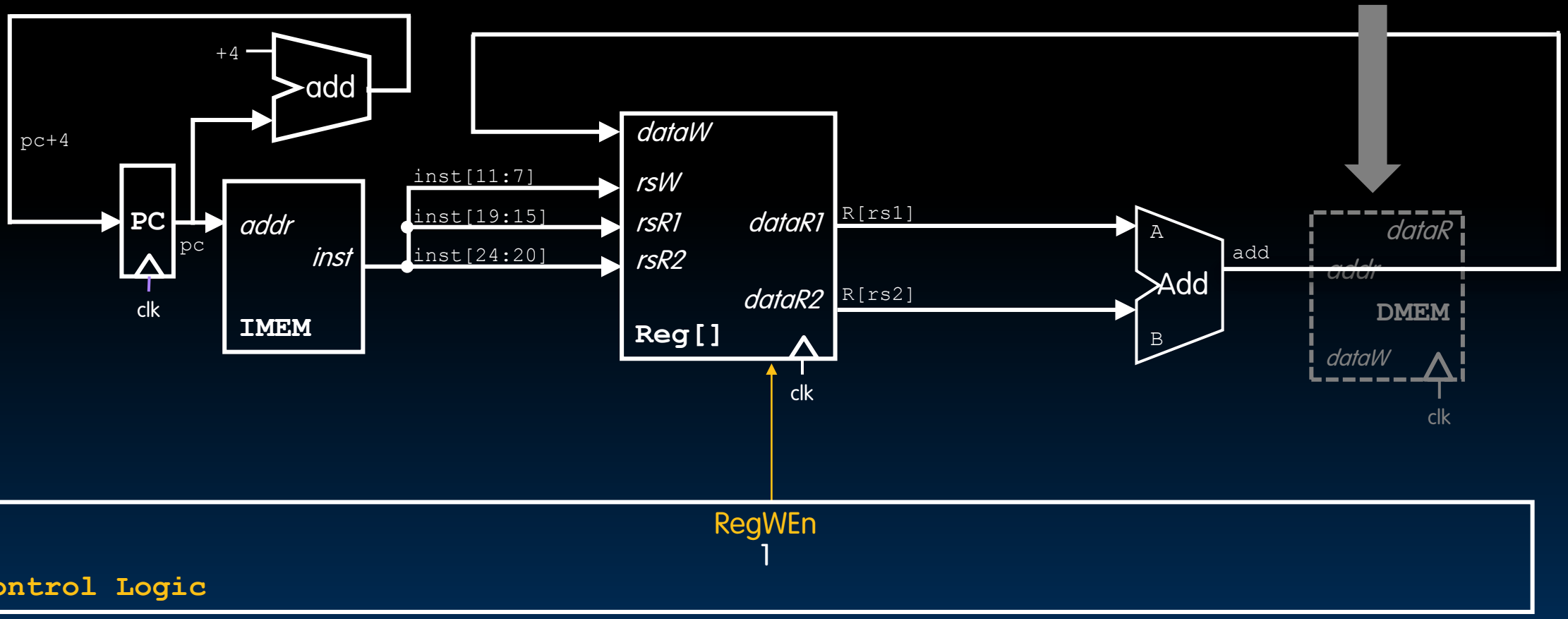
Write Add output to destination register.



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Plan: Iteratively Build Full Datapath

For now, we will disconnect memory.
By the end of next lecture, we'll build the datapath that supports the full RV32I ISA.



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R-Type: `sub` Datapath

- Building a RISC-V Processor
- CPU Elements and Stages
- R-Type: **`add`** datapath
- R-Type: **`sub`** datapath
- Datapath with immediates: **`addi`**

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Implementing the `sub` Instruction

- What if our ISA now had two instructions: add/sub?

funct7			funct3		opcode	
0000000	rs2	rs1	000	rd	0110011	add
0100000	rs2	rs1	000	rd	0110011	sub

```
sub  rd, rs1, rs2
```

- `sub` is almost the same as `add`, except now the ALU subtracts operands instead of adding them:
  - RegFile `Reg[rd] = Reg[rs1] - Reg[rs2]`
  - PC `PC = PC + 4`
- Instruction bit `inst[30]` selects between `add`/`sub`.
  - Details left to *control logic*.

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Datapath for sub

31 25	24 20	19 15	14 12	11 7	6 0
0100000	rs2	rs1	funct3	rd	0110011

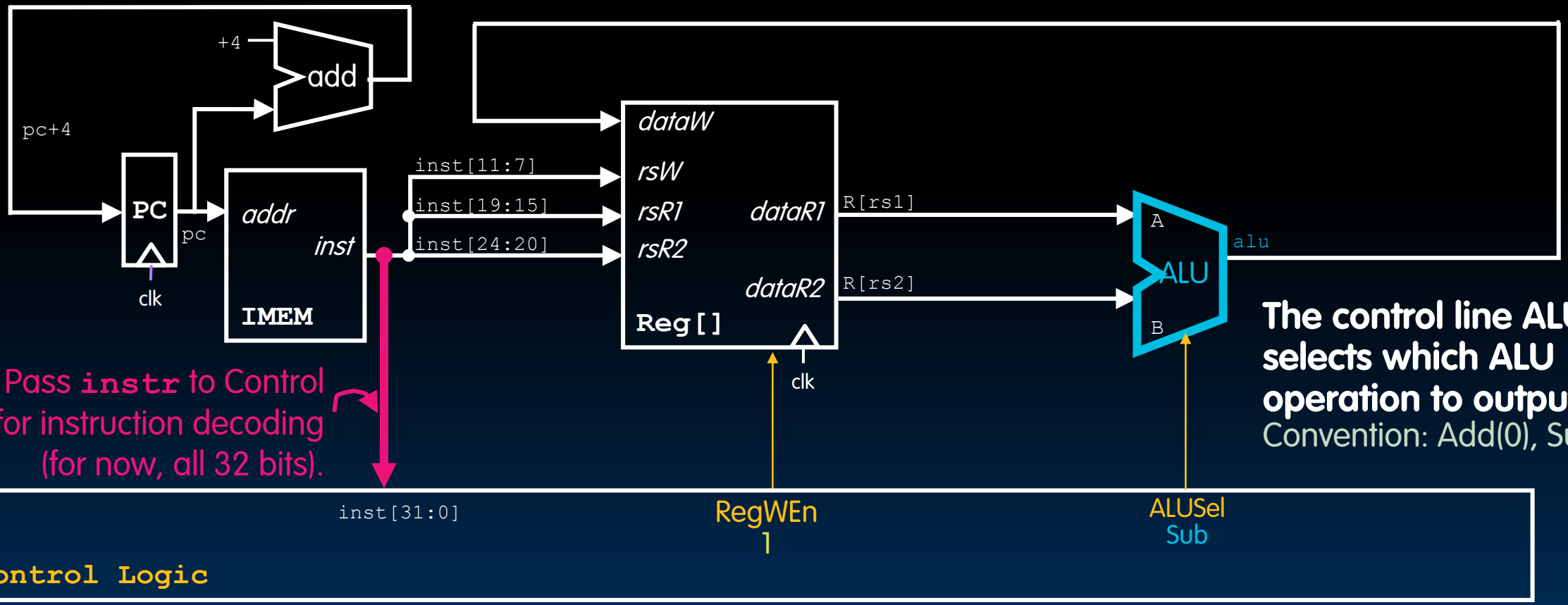
```
PC = PC + 4        R[rd] = R[rs1] - R[rs2]
```

Increment PC to next instruction.

Split instruction to index into RegFile.

Feed read register values *into ALU.*

Write *ALU output* to destination register.



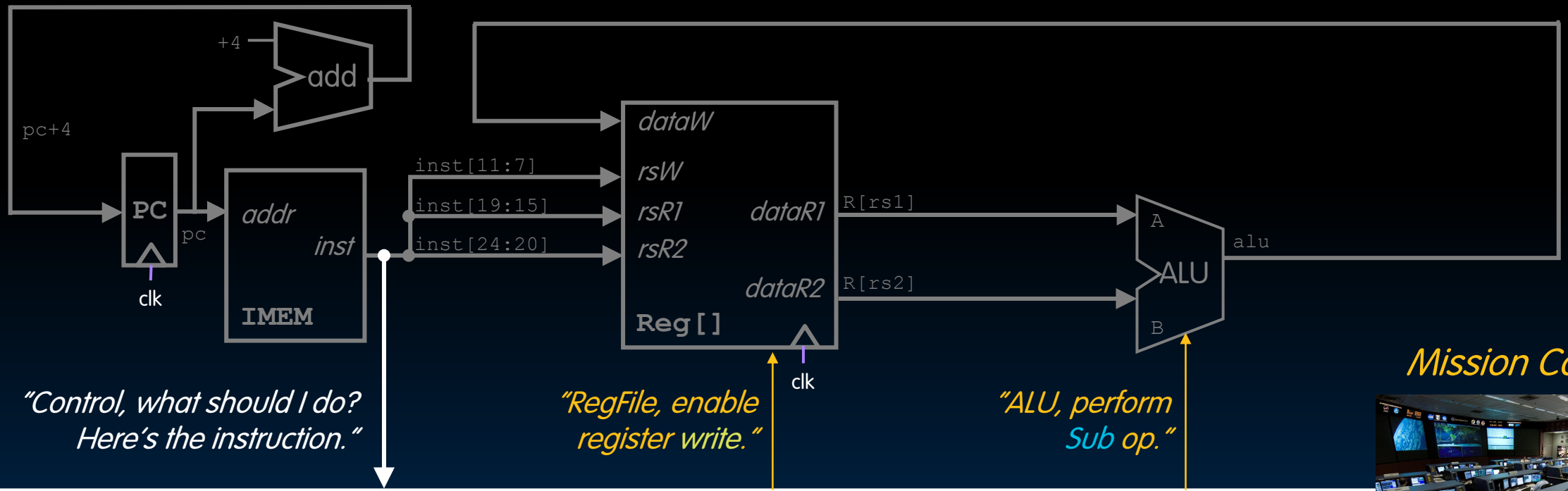
The control line ALUSel selects which ALU operation to output.
Convention: Add(0), Sub(1).

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Control vs. Datapath Analogy: Space Mission

Control and Datapath are two *abstractions* that together comprise the CPU. Under the hood, both are designed with combinational logic and/or state elements.





Mission Control



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Supporting all R-Type Instructions

funct7			funct3		opcode	
0000000	rs2	rs1	000	rd	0110011	add
0100000	rs2	rs1	000	rd	0110011	sub
0000000	rs2	rs1	001	rd	0110011	sll
0000000	rs2	rs1	010	rd	0110011	slt
0000000	rs2	rs1	011	rd	0110011	sltu
0000000	rs2	rs1	100	rd	0110011	xor
0000000	rs2	rs1	101	rd	0110011	srl
0100000	rs2	rs1	101	rd	0110011	sra
0000000	rs2	rs1	110	rd	0110011	or
0000000	rs2	rs1	111	rd	0110011	and



ALUSel
add,sub,xor,and,or,
slt,sltu,sll,sra,srl

The Control Logic decodes **funct3**, **funct7** instruction fields and selects appropriate ALU function by setting the control line ALUSel.

Check out Project 3 for the full ALU!

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Datapath with Immediates: `addi`

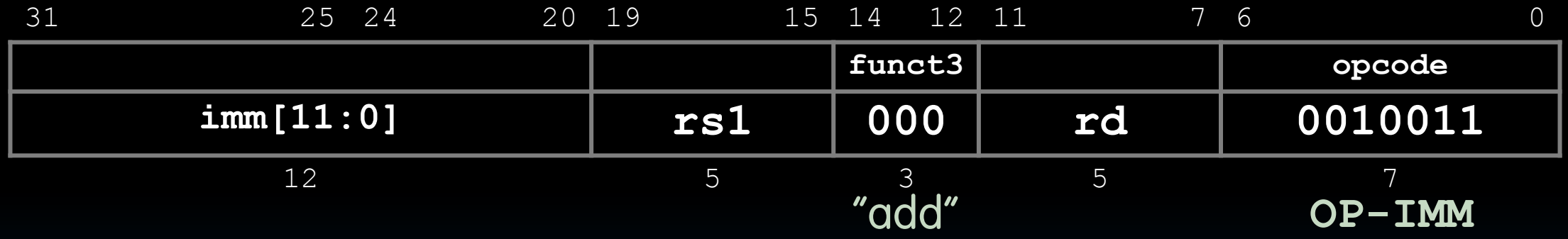
- Building a RISC-V Processor
- CPU Elements and Stages
- R-Type: **`add`** datapath
- R-Type: **`sub`** datapath
- Datapath with immediates: **`addi`**

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Implementing the addi Instruction

- Suppose we add a new instruction: addi, RV32I I-Format:

```
addi   rd, rs1, imm
```



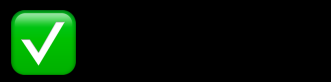
- The addi instruction updates the same two states as before. But we now need to build an immediate imm!
  - RegFile
  - PC

```
Reg[rd] = Reg[rs1] + imm
PC = PC + 4
```


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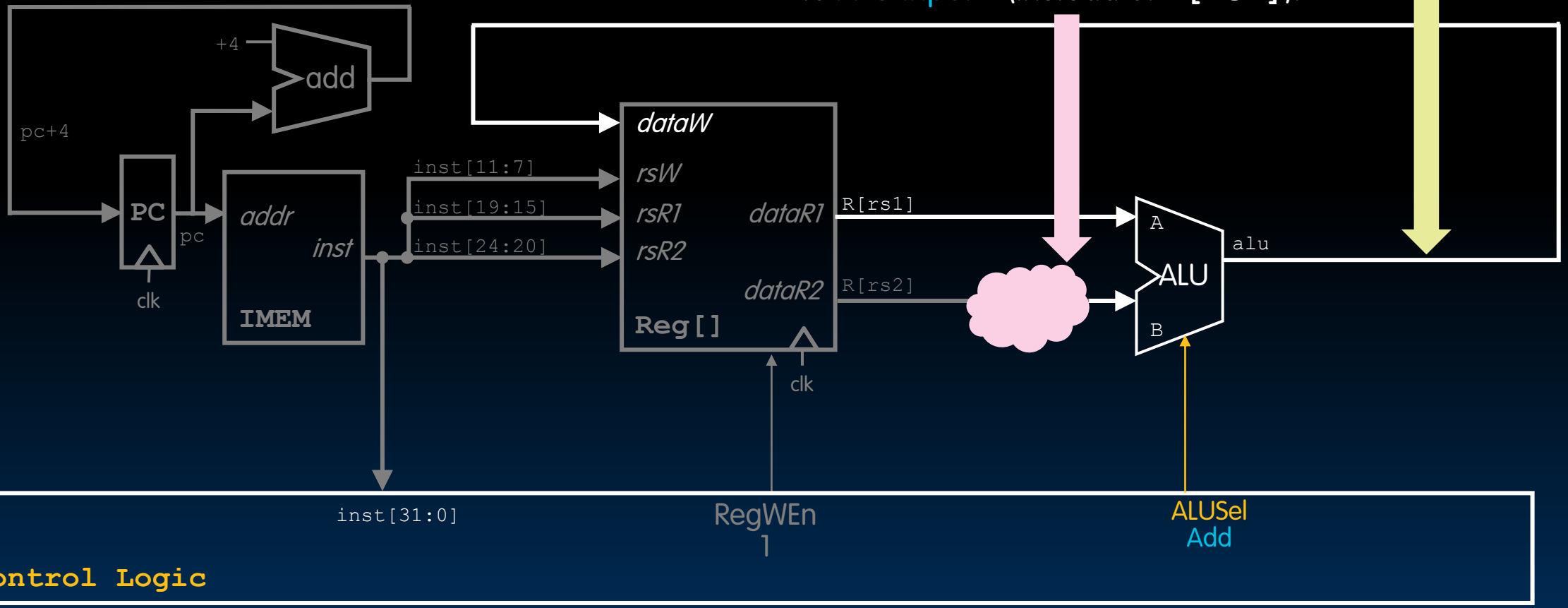
What Does the Datapath Need for `addi`?

`PC = PC + 4` `R[rd] = R[rs1] + imm`



To compute `alu = R[rs1] + imm`…

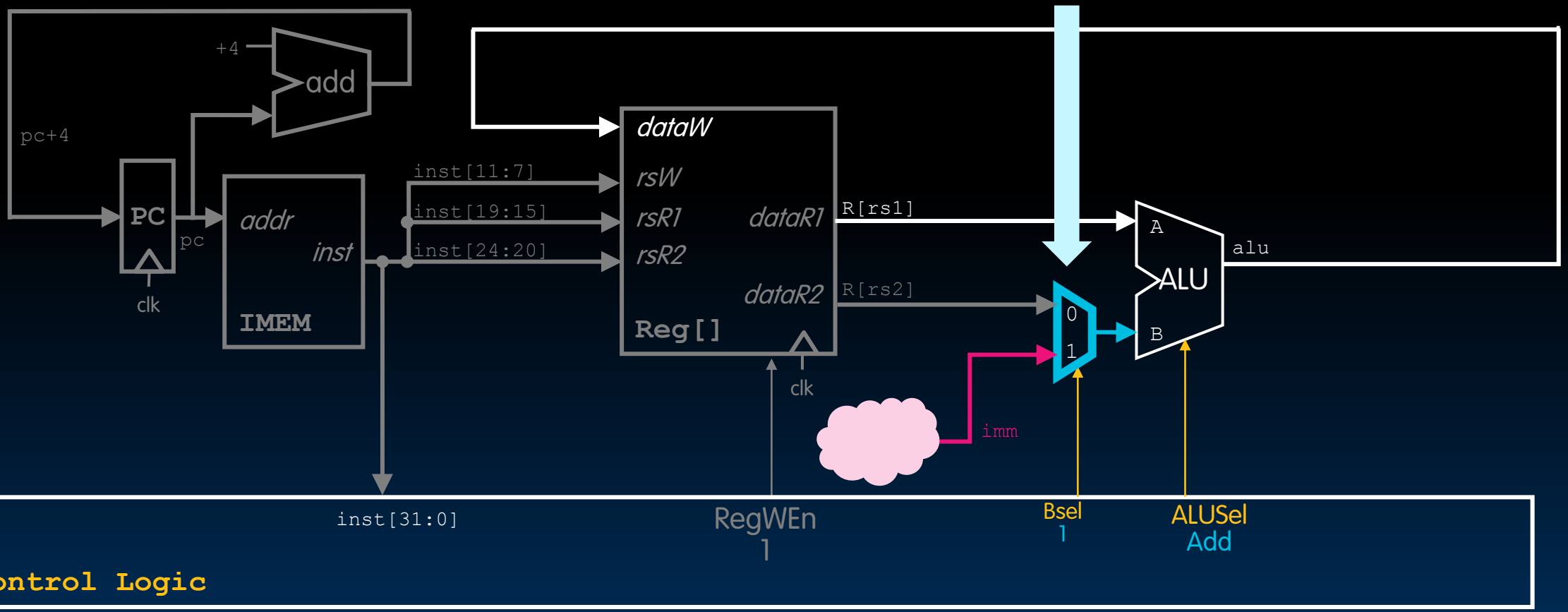
…we should feed in an *immediate* to ALU input B (instead of `R[rs2]`).



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1. New MUX to Select Immediate for ALU

1. Control line Bsel=1 selects the generated immediate **imm** for ALU input B.

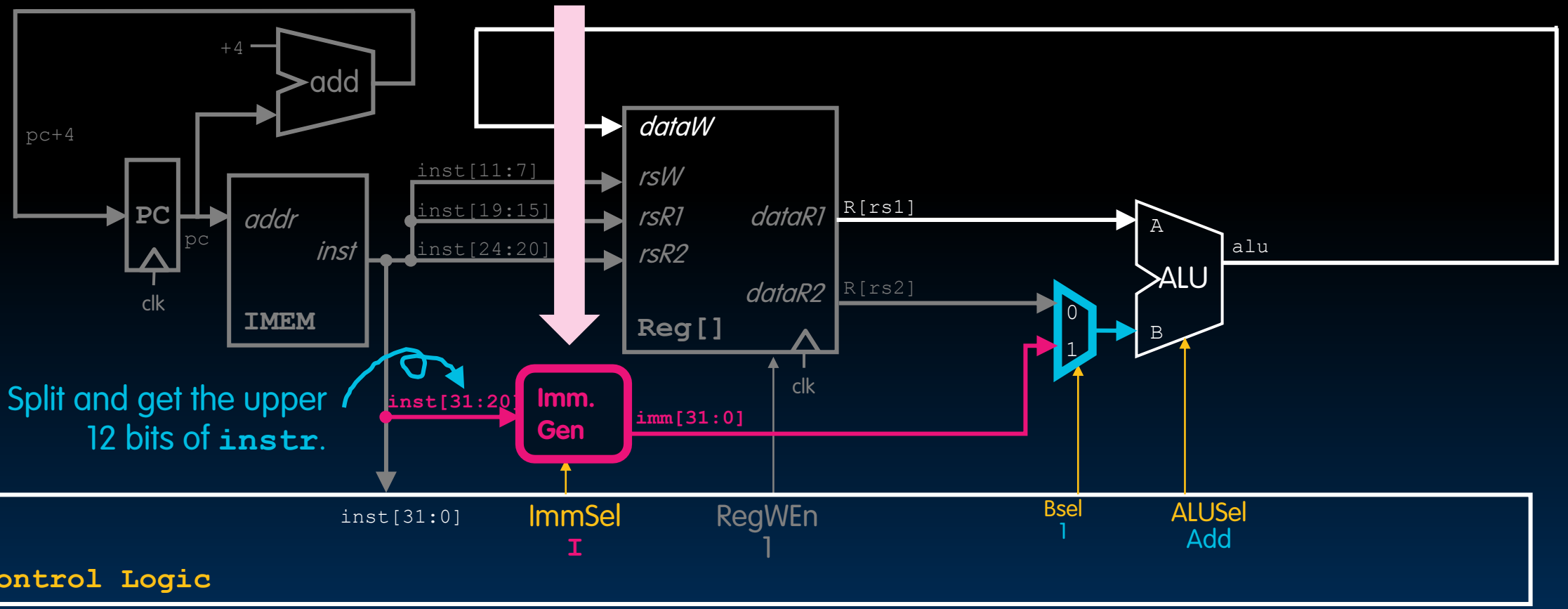


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2. New Block to Generate 32-bit Immediate

2. *Immediate Generation Block* builds a 32-bit immediate **imm** from instruction bits.

1. Control line Bsel=1 selects the generated immediate **imm** for ALU input B.



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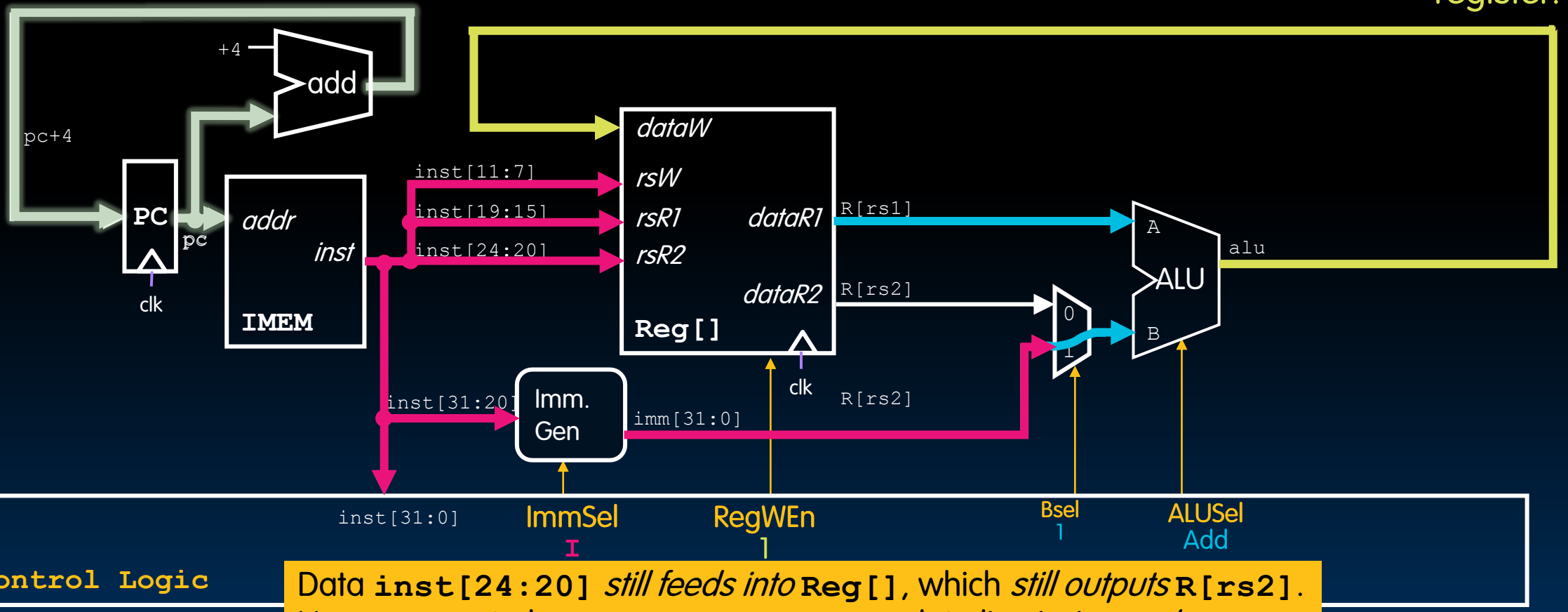
Tracing/Lighting the `addi` Datapath

Increment PC to next instruction.

Immediate Generation Block builds a 32-bit immediate **`imm`** from instruction bits.

Control line Bsel=1 selects the generated immediate **`imm`** for ALU input B.

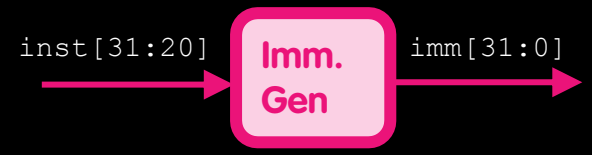
Write ALU output to destination register.



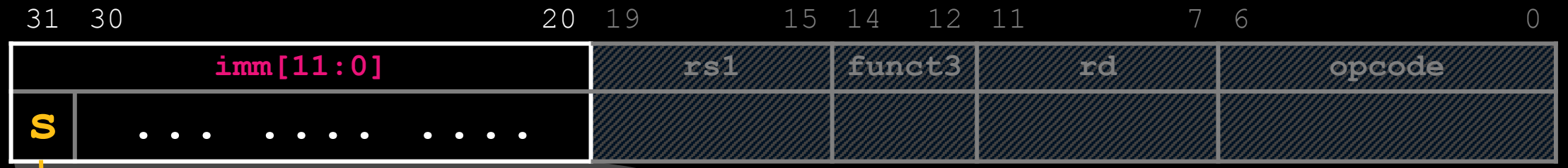
Data **`inst[24:20]`** *still feeds into* **`Reg[]`**, which *still outputs* **`R[rs2]`**. However, control **`Bsel=1`** means **`R[rs2]`** data line is *ignored*.

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Immediate Generation Block Design (more next time)



Instruction
`inst[31:0]`



Copy *upper* 12 bits of instruction, `inst[31:20]`, to *lower* 12 bits of immediate, `imm[11:0]`.

Immediate
`imm[31:0]`

Sign-extend: Copy `inst[31]` to upper 20 bits of immediate, `imm[31:12]`.



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Datapath Design I

Your Own Design

1. We should use the main ALU to compute `PC=PC+4` in order to save some gates.
2. The ALU is a synchronous state element.
3. The program counter (PC) is a register.

FFF

FFT

FTF

FTT

TFF

TFT

TTF

TTT

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Your Own Design

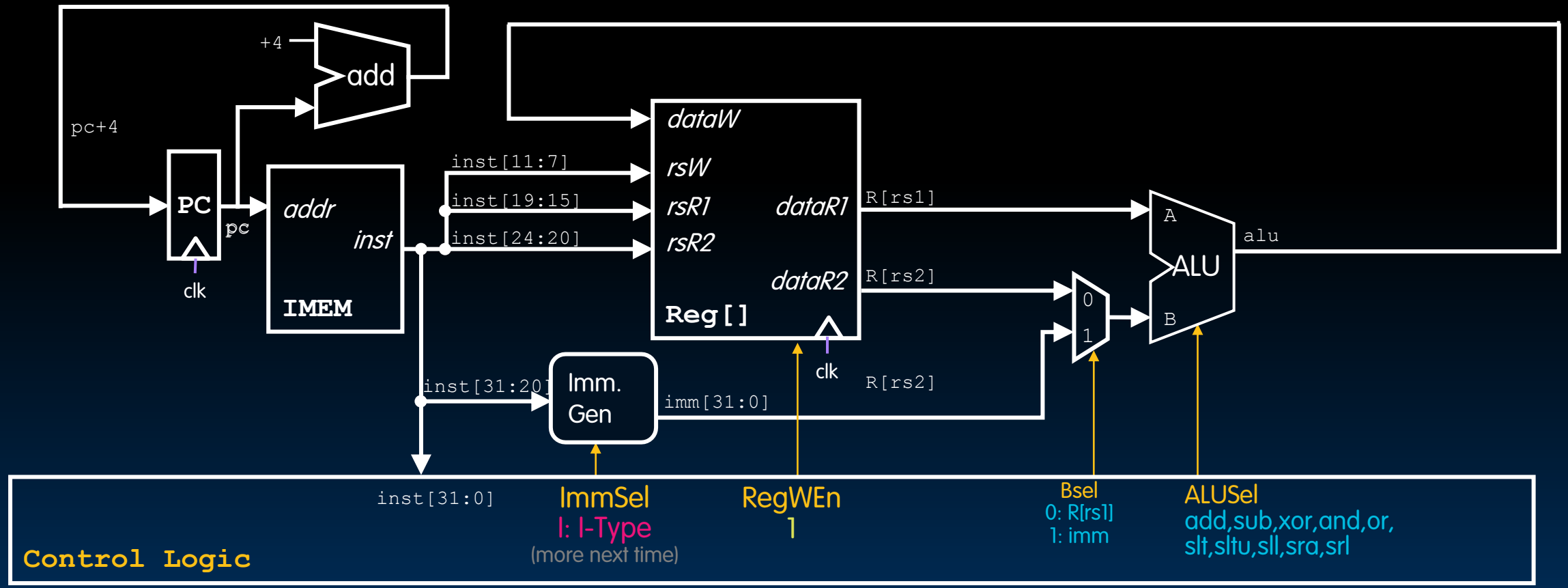
1. We should use the main ALU to compute `PC=PC+4` in order to save some gates.
2. The ALU is a synchronous state element.
3. The program counter (PC) is a register.

	123
A.	FFF
B.	FFT
C.	FTF
D.	FTT
E.	TFF
F.	TFT
G.	TTF
H.	TTT

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Summary: Arithmetic/Logical Datapath

- All data lines carry information.
- Control logic determines what is "useful/needed" vs. what is "ignored."
  - e.g., ALUSel: chooses ALU operation; Bsel: chooses register/immediate for ALU input B.



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